

SOLIDWORKS API (.NET VBA C#) — Reference (220/337): IPartDoc (part 3 of 4)

SOLIDWORKS API Documentation (converted from CHM)

IPartDoc (part 3 of 4) (Type)

SOLIDWORKS API Help

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IGetEntityByName Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetEntityByName Method (IPartDoc)

Gets an entity (face, edge, vertex) by name.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEntityByName( _  
    ByVal Name As System.String, _  
    ByVal EntityType As System.Integer _  
) As Entity
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Name As System.String  
Dim EntityType As System.Integer  
Dim value As Entity  
  
value = instance.IGetEntityByName(Name, EntityType)
```

C#

```
Entity IGetEntityByName(  
    System.string Name,  
    System.int EntityType  
)
```

C++/CLI

```
Entity^ IGetEntityByName(  
& System.String^ Name,  
& System.int EntityType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Name of entity

EntityType

Entity type as defined in [swSelectType_e](#) (see **Remarks**)

Return Value

[Entity](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IGetEntityByName](#).

Example

[Get Edge Data By Name \(C++\)](#)

Remarks

Valid entityType types:

- swSelFACES
- swSelEDGES

- swSeleVERTICES

Another way to obtain an [IEntity](#) object based on its name is to traverse the body topology (for example, use [IBody2::GetFirstFace](#) or [IBody2::IGetFirstFace](#) and so on) and use [IModelDoc2::GetEntityName](#) or [IModelDoc2::IGetEntityName](#) to determine when you have located the desired entity.

Because the names of entities are stored with the part document, this method resides on the [IPartDoc](#) object.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetEntityByName Method](#)
[IPartDoc::GetEntityName Method](#)
[IPartDoc::DeleteEntityName Method](#)
[IPartDoc::GetNamedEntities Method](#)
[IPartDoc::GetNamedEntities Method](#)
[IPartDoc::IDeleteEntityName Method](#)
[IPartDoc::IGetEntityByName Method](#)
[IPartDoc::IGetEntityName Method](#)
[IPartDoc::IGetNamedEntities Method](#)
[IPartDoc::SetEntityName Method](#)

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IGetEntityName Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetEntityName Method (IPartDoc)

Gets the name of the specified entity.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEntityName( _
    ByVal Entity As Entity _
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim Entity As Entity
Dim value As System.String

value = instance.IGetEntityName(Entity)
```

C#

```
System.string IGetEntityName(
    Entity Entity
)
```

C++/CLI

```
System.String^ IGetEntityName(
    & Entity^ Entity
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entity
[Entity](#)

Return Value

Name of entity

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IGetEntityName](#).

Example

Remarks

Valid for faces, edges, and vertices only.

SOLIDWORKS suggests that you use the more comprehensive method [IModelDoc2::GetEntityName](#) or [IModelDoc2::IGetEntityName](#) instead of this method.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::DeleteEntityName Method](#)
[IPartDoc::GetEntityByName Method](#)
[IPartDoc::GetEntityName Method](#)
[IPartDoc::GetNamedEntities Method](#)
[IPartDoc::GetNamedEntitiesCount Method](#)
[IPartDoc::IDeleteEntityName Method](#)
[IPartDoc::IGetEntityByName Method](#)
[IPartDoc::IGetEntityName Method](#)
[IPartDoc::IGetNamedEntities Method](#)
[IPartDoc::SetEntityName Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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IGetNamedEntities Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetNamedEntities Method (IPartDoc)

Gets an array of names of named entities (face, edge, vertex, and so on) in this part.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetNamedEntities( _  
    ByVal Count As System.Integer _  
) As Entity
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Count As System.Integer  
Dim value As Entity  
  
value = instance.IGetNamedEntities(Count)
```

C#

```
Entity IGetNamedEntities(  
    System.int Count  
)
```

C++/CLI

```
Entity^ IGetNamedEntities(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of named entities

Return Value

- in-process, unmanaged C++: Pointer to an array of named [entities](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Example

[Get Named Entities \(C++\)](#)

Remarks

Call [IPartDoc::GetNamedEntitiesCount](#) before calling this method to get Count.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetEntityByName Method](#)
[IPartDoc::GetEntityName Method](#)
[IPartDoc::GetNamedEntities Method](#)
[IPartDoc::GetNamedEntitiesCount Method](#)
[IPartDoc::DeleteEntityName Method](#)
[IPartDoc::IGetEntityByName Method](#)
[IPartDoc::IGetEntityName Method](#)
[IPartDoc::IGetNamedEntities Method](#)
[IPartDoc::IDeleteEntityName Method](#)
[IPartDoc::SetEntityName Method](#)

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IGetPartBox Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetPartBox Method (IPartDoc)

Gets the box enclosing the part.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPartBox( _  
    ByVal NoConversion As System.Boolean _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim NoConversion As System.Boolean  
Dim value As System.Double  
  
value = instance.IGetPartBox(NoConversion)
```

C#

```
System.double IGetPartBox(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.double IGetPartBox(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

Convert to user units or not; True retains system units, false changes to user units

Return Value

- in-process, unmanaged C++: Pointer to an array of 6 doubles containing the 2 diagonal points that bound the part
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Example

[Return Box Extents \(C++\)](#)

Remarks

The values returned are approximate and should not be used for comparison or calculation purposes. Furthermore, the bounding box may vary after rebuilding the model.

The X,Y,Z points returned are the lower- and upper-diagonal corners that bound the part with the box sides parallel to the X, Y and Z axes. The box dimensions returned enclose the part and usually close to the minimum possible size, but this is not guaranteed.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

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IGetProcessedBody Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetProcessedBody Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::IGetProcessedBody2](#).

.NET Syntax

Visual Basic (Declaration)

Function IGetProcessedBody() As Body

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As Body

value = instance.IGetProcessedBody()
```

C#

Body IGetProcessedBody()

C++/CLI

Body^ IGetProcessedBody();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IGetProcessedBody](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

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IGetProcessedBody2 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetProcessedBody2 Method (IPartDoc)

Pre-processes the geometry of a body so that:

- Closed periodic faces (for example, the lateral face of a cylinder) are split into two faces.
- Faces that straddle the seam, if any, of the underlying surface are split into two faces.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetProcessedBody2() As Body2
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As Body2  
  
value = instance.IGetProcessedBody2()
```

C#

```
Body2 IGetProcessedBody2()
```

C++/CLI

```
Body2^ IGetProcessedBody2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Processed [body](#), which is a copy of the body for this part

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IGetProcessedBody2](#).

Remarks

Preprocessing is sometimes necessary (for example, IGES files), for exporting to systems that have difficulty with periodic conditions.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IBody2::GetProcessedBody2 Method](#)
[IBody2::GetProcessedBodyWithSelFace Method](#)
[IBody2::IGetProcessedBodyWithSelFace Method](#)
[IPartDoc::IGetProcessedBodyWithSelFace2 Method](#)
[IBody2::IGetProcessedBody Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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IGetProcessedBodyWithSelFace Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetProcessedBodyWithSelFace Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::IGetProcessedBodyWithSelFace2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetProcessedBodyWithSelFace() As Body
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As Body  
  
value = instance.IGetProcessedBodyWithSelFace()
```

C#

```
Body IGetProcessedBodyWithSelFace()
```

C++/CLI

```
Body^ IGetProcessedBodyWithSelFace();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IGetProcessedBodyWithSelFace](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

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IGetProcessedBodyWithSelFace2 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetProcessedBodyWithSelFace2 Method (IPartDoc)

Gets a processed body.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetProcessedBodyWithSelFace2() As Body2
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As Body2

value = instance.IGetProcessedBodyWithSelFace2()
```

C#

```
Body2 IGetProcessedBodyWithSelFace2()
```

C++/CLI

```
Body2^ IGetProcessedBodyWithSelFace2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Processed [body](#), which is a copy of the body for this part

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IGetProcessedBodyWithSelFace2](#).

Remarks

This method requires that you preselect a face.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::IGetProcessedBody2 Method](#)
[IBody2::GetProcessedBody2 Method](#)
[IBody2::GetProcessedBodyWithSelFace Method](#)
[IBody2::IGetProcessedBody Method](#)
[IBody2::IGetProcessedBodyWithSelFace Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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IGetSectionedBody Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetSectionedBody Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::IGetSectionedBody2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSectionedBody( _  
    ByVal ViewIn As ModelView _  
) As Body
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim ViewIn As ModelView  
Dim value As Body  
  
value = instance.IGetSectionedBody(ViewIn)
```

C#

```
Body IGetSectionedBody(  
    ModelView ViewIn  
)
```

C++/CLI

```
Body^ IGetSectionedBody(  
& ModelView^ ViewIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewIn

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IGetSectionedBody](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

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IGetSectionedBody2 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetSectionedBody2 Method (IPartDoc)

Gets the sectioned body seen in the specified drawing view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSectionedBody2( _  
    ByVal ViewIn As ModelView _
```

) As Body2

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim ViewIn As ModelView  
Dim value As Body2  
  
value = instance.IGetSectionedBody2(ViewIn)
```

C#

```
Body2 IGetSectionedBody2(  
    ModelView ViewIn  
)
```

C++/CLI

```
Body2^ IGetSectionedBody2(  
& ModelView^ ViewIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewIn
[Model view](#)

Return Value

[Sectioned body](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IGetSectionedBody2](#).

Remarks

To determine if the desired model view is currently displaying a sectioned view, use [IModelView::GetDisplayState](#).

If you need the full body representation, use [IPartDoc::GetBodies2](#) or [IPartDoc::EnumBodies3](#), which ignores the sectioning operation.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::IGetSectionedBody2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

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IGetTessNorms Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetTessNorms Method (IPartDoc)

Gets the normal vector for each of the triangles, which make up the shaded picture tessellation.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessNorms() As System.Single
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.Single

value = instance.IGetTessNorms()
```

C#

```
System.float IGetTessNorms()
```

C++/CLI

```
System.float IGetTessNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of floats (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The total size of the data is:

[9 x *sizeof(float)* x (*number of triangles*)].

The format of the returned data is:

- *float* x, y, z first point's unit normal
- *float* x, y, z second point's unit normal
- *float* x, y, z third point's unit normal

for the set of triangles for the part.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)

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IGetTessTriangles Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetTessTriangles Method (IPartDoc)

Gets the triangles that make up the shaded picture tessellation for this part.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriangles( _  
    ByVal NoConversion As System.Boolean _  
) As System.Single
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim NoConversion As System.Boolean  
Dim value As System.Single
```

```
value = instance.IGetTessTriangles(NoConversion)
```

C#

```
System.float IGetTessTriangles(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.float IGetTessTriangles(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True to prohibit conversion to user units from system units , false to do so

Return Value

- in-process, unmanaged C++: Pointer to an array of floats (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

These triangles are intended for graphics display purposes and do not represent a tessellation that could be used, for example, by a machining application. If you need the type of accuracy associated with a machining product, it is recommended that you traverse the body faces and extract the topology and geometry data to create your own faceting.

The total size of the data is:

[9 x sizeof(float) x (number of triangles)].

The format of the data is:

- *float* x, y, z first point in meters
- *float* x, y, z second point in meters
- *float* x, y, z third point in meters

for the set of triangles for the part.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[ITessellation Interface](#)
[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)

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IGetTessTriStripEdges Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetTessTriStripEdges Method (IPartDoc)

Gets the edge ID for each of the edges in the triangles that make up the tessellation for this part.

.NET Syntax

Visual Basic (Declaration)

Function IGetTessTriStripEdges() As System.Integer

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.Integer

value = instance.IGetTessTriStripEdges()
```

C#

```
System.int IGetTessTriStripEdges()
```

C++/CLI

```
System.int IGetTessTriStripEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of longs containing the list of edge IDs for this part document (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The format of the returned data is:

DWORD <i>FaceCount</i>	
DWORD <i>StripCount</i>	
DWORD <i>EdgeCount</i>	
ForEach <i>Face</i>	
DWORD <i>StripCount</i>	
DWORD [<i>StripLengths</i>]	
ForEach <i>Strip</i>	(where this is an array of edge IDs for each strip from 1 to (VertexPerStrip-1). Element 0 indicates if that a triangle strip exists (value = 1) or a single triangle (value = 0))
Long [<i>Edgels</i>]	

where:

<i>FaceCount</i> :	Number of faces on the body
<i>StripCount</i> :	Total number of strips on the body
<i>EdgeCount</i> :	Total number of vertices
<i>NumStrips</i> :	Number of strips on a particular face
<i>StripLengths</i> :	Array containing the number of vertex points on particular face strip
<i>Edgels</i> :	Array of edge IDs for each vertex on the particular face strip

For triangle strips, *Edgels* is:

If the strip has n vertices, then there are $2(n-2) + 1$ edge tags. The i -th element of the tags array (starting at $i = 1$) will be the tag of the edge between vertex $\text{floor}((i-1)/2) + 1$ and vertex $\text{floor}(i/2) + 2$, where $\text{floor}(x)$ is the integer portion of x .

For single triangles, *Edgels* is:

The i -th element of the tags array (starting at $i = 1$) will be the tag of the edge between vertex $i-1$ (if $i=1$ then $i=n$) and vertex i .

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)

Availability

SOLIDWORKS 99 SP2, datecode 1999250

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IGetTessTriStripEdgeSize Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetTessTriStripEdgeSize Method (IPartDoc)

Gets the size of the array returned by [IPartDoc::GetTessTriStripEdges](#) and [IPartDoc::IGetTessTriStripEdges](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriStripEdgeSize() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.Integer

value = instance.IGetTessTriStripEdgeSize()
```

C#

```
System.int IGetTessTriStripEdgeSize()
```

C++/CLI

```
System.int IGetTessTriStripEdgeSize();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Size of the array returned by [IPartDoc::GetTessTriStripEdges](#) and [IPartDoc::IGetTessTriStripEdges](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IGetTessTriStripEdgeSize](#).

Remarks

The number is calculated as (3 + *FaceCount* + *StripCount* + *EdgeCount*).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)

Availability

SOLIDWORKS 99 SP2, datecode 1999250

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IGetTessTriStripNorms Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetTessTriStripNorms Method (IPartDoc)

Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for this part.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriStripNorms() As System.Single
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.Single
```

```
value = instance.IGetTessTriStripNorms()
```

C#

```
System.float IGetTessTriStripNorms()
```

C++/CLI

```
System.float IGetTessTriStripNorms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array containing the tristrip normals (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The format of the returned data is:

DWORD FaceCount

DWORD StripCount

DWORD (VertexCount) x 3

DWORD NumStrips

DWORD [VertexPerStrip] = array from 0 to (*Numstrips*-1).

Float [Normals] = array of X,Y,Z normal components for each strip from 0 to (*VertexPerStrip*-1).

where

FaceCount = Number of faces on the body.

StripCount = Total number of strips on the body.

VertexCount x 3 = Total number of vertices. Multiplied by 3 to cover X, Y, and Z.

NumStrips = Number of strips on a particular face.

VertexPerStrip = Array containing the number of vertex points on particular face strip.

NormalComp = Array of X,Y,Z normal components for each vertex on the particular face strip.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)

[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStrips Method](#)
[ITessellation Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTessTriStrips Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IGetTessTriStrips Method (IPartDoc)

Gets the vertices that make up the shaded picture tessellation for this part.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTessTriStrips( _  
    ByVal NoConversion As System.Boolean _  
) As System.Single
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim NoConversion As System.Boolean  
Dim value As System.Single  
  
value = instance.IGetTessTriStrips(NoConversion)
```

C#

```
System.float IGetTessTriStrips(  
    System.bool NoConversion  
)
```

C++/CLI

```
System.float IGetTessTriStrips(  
& System.bool NoConversion  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NoConversion

True prohibits converting to user units from system units, false does not

Return Value

- in-process, unmanaged C++: Pointer to an array of floats (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The vertices are all unique for that strip. In other words, a vertex that is shared by two tessellation triangles will only appear once in the return value.

These triangles are intended for graphics display purposes and do not represent a tessellation that could be used, for example, by a machining application. If you need the type of accuracy associated with a machining product, it is recommended that you traverse the body faces and extract the topology and geometry data to create your own faceting.

These triangles are intended for graphics display purposes and do not represent a tessellation that can be used, for example, by a machining application. If you need the level of accuracy associated with a machining product, traverse the body faces and extract the topology and geometry data to create your own faceting.

This method returns an array of float values in the form of:

[*NumStrips*,

VertexPerStrip1, *VertexPerStrip2*,..., *VertexPerStripN*,

VertexPoints1[*VertexPerStrip1* elements]*3,
VertexPoints2[*VertexPerStrip2* elements]*3,...,
VertexPointsN[*VertexPerStripN* elements]*3]

where:

NumStrips = number of triangle strips on a particular face.

VertexPerStripN = number of vertex points for the *N*th triangle strip.

VertexPointsN = a sub-array of vertices consisting of three float values representing the x,y,z coordinate of the vertex; the sub-array's length is defined by the *VertexPerStripN* element in the array.

Because the values *NumStrips* and the elements of *VertexPerStrip* are long values packed into single values, you must [unpack](#) them before they can be used.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetTessNorms Method](#)
[IPartDoc::GetTessTriangleCount Method](#)
[IPartDoc::GetTessTriangles Method](#)
[IPartDoc::GetTessTriStripEdges Method](#)
[IPartDoc::GetTessTriStripNorms Method](#)
[IPartDoc::GetTessTriStrips Method](#)
[IPartDoc::GetTessTriStripSize Method](#)
[IPartDoc::IGetTessNorms Method](#)
[IPartDoc::IGetTessTriangles Method](#)
[IPartDoc::IGetTessTriStripEdges Method](#)
[IPartDoc::IGetTessTriStripEdgeSize Method](#)
[IPartDoc::IGetTessTriStripNorms Method](#)
[ITessellation Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMaterialPropertyValues Property (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IMaterialPropertyValues Property (IPartDoc)

Gets or sets a material's properties in the active configuration.

.NET Syntax

Visual Basic (Declaration)

Property IMaterialPropertyValues As System.Double

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.Double  
  
instance.IMaterialPropertyValues = value  
value = instance.IMaterialPropertyValues
```

C#

```
System.double IMaterialPropertyValues {get; set;}
```

C++/CLI

```
property System.double IMaterialPropertyValues {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles describing the material's values (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IMaterialPropertyValues](#).

Remarks

The material values include the color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency and emission.

The format of the parameters or return values is an array of doubles as follows:

[*R, G, B, Ambient, Diffuse, Specular, Shininess, Transparency, Emission*]

To see a color change, you must:

1. Specify *R*, *G*, and *B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse*, *Specular*, *Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient*, *Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after setting this property.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::MaterialPropertyValues Property](#)
[IPartDoc::MaterialUserName Property](#)
[IPartDoc::MaterialIdName Property](#)
[IPartDoc::SetMaterialVisualProperties Method](#)
[IPartDoc::SetMaterialPropertyName2 Method](#)
[IPartDoc::GetMaterialVisualProperties Method](#)
[IPartDoc::GetMaterialPropertyName2 Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ImportDiagnosis Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ImportDiagnosis Method (IPartDoc)

Diagnoses and repairs any gaps or bad faces on imported features.

VB.NET Syntax

Visual Basic (Declaration)

```
Function ImportDiagnosis( _  
    ByVal CloseAllGaps As System.Boolean, _  
    ByVal RemoveFaces As System.Boolean, _  
    ByVal FixFaces As System.Boolean, _  
    ByVal Options As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim CloseAllGaps As System.Boolean  
Dim RemoveFaces As System.Boolean  
Dim FixFaces As System.Boolean  
Dim Options As System.Integer  
Dim value As System.Integer  
  
value = instance.ImportDiagnosis(CloseAllGaps, RemoveFaces, FixFaces, Options)
```

C#

```
System.int ImportDiagnosis(  
    System.bool CloseAllGaps,  
    System.bool RemoveFaces,  
    System.bool FixFaces,  
    System.int Options  
)
```

C++/CLI

```
System.int ImportDiagnosis(  
& System.bool CloseAllGaps,  
& System.bool RemoveFaces,  
& System.bool FixFaces,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CloseAllGaps

True to repair any gaps, false to not

RemoveFaces

True to remove any bad faces and create gaps in the feature, false to not

FixFaces

True to fix the bad faces, false to not

Options

Not used

Return Value

>= 0 if import diagnosis is successful, -1 if an error occurred

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ImportDiagnosis](#).

Example

[Import STEP File \(C#\)](#)

[Import STEP File \(VB.NET\)](#)

[Import STEP File \(VBA\)](#)

Remarks

Use this method for an imported solid body that has rebuild errors or for an imported surface that did not knit into a solid body.

See Also

[IPartDoc Interface](#)

[IPartDoc Members](#)

[IPartDoc::ImportDiagnosisGapCloser Method](#)

[iBody2::Diagnose Method](#)

[IDiagnoseResult Interface](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

ImportDiagnosisGapCloser Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ImportDiagnosisGapCloser Method (IPartDoc)

Allows you to repair a gap by moving the vertices on the edges surrounding the gap to new positions to close the gap on the imported model.

.NET Syntax

Visual Basic (Declaration)

```
Sub ImportDiagnosisGapCloser( _  
    ByVal OldX As System.Double, _  
    ByVal OldY As System.Double, _  
    ByVal OldZ As System.Double, _  
    ByVal NewX As System.Double, _  
    ByVal NewY As System.Double, _  
    ByVal NewZ As System.Double, _  
    ByVal LastMove As System.Boolean _  
)
```

Visual Basic (Usage)

```
Dim instance As IPartDoc  
Dim OldX As System.Double  
Dim OldY As System.Double
```



```
Dim OldZ As System.Double
Dim NewX As System.Double
Dim NewY As System.Double
Dim NewZ As System.Double
Dim LastMove As System.Boolean
```

```
instance.ImportDiagnosisGapCloser(OldX, OldY, OldZ, NewX, NewY, NewZ, LastMove)
```

C#

```
void ImportDiagnosisGapCloser(
    System.double OldX,
    System.double OldY,
    System.double OldZ,
    System.double NewX,
    System.double NewY,
    System.double NewZ,
    System.bool LastMove
)
```

C++/CLI

```
void ImportDiagnosisGapCloser(
    & System.double OldX,
    & System.double OldY,
    & System.double OldZ,
    & System.double NewX,
    & System.double NewY,
    & System.double NewZ,
    & System.bool LastMove
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OldX

x coordinate of vertex to move

OldY

y coordinate of vertex to move

OldZ

z coordinate of vertex to move

NewX

x coordinate where to move the vertex

NewY

y coordinate where to move the vertex

NewZ

z coordinate where to move the vertex

LastMove

True if this move is the last move in a series of moves to close the gap, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ImportDiagnosisGapCloser](#).

See Also

[IPartDoc Interface](#)

[IPartDoc Members](#)

[IPartDoc::ImportDiagnosis Method](#)

[IBody2::Diagnose Method](#)

[IDiagnoseResult Interface](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

Obsolete. Superseded by [IPartDoc::InsertPart](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBasePart( _  
    ByVal FileName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim FileName As System.String  
Dim value As System.Boolean  
  
value = instance.InsertBasePart(FileName)
```

C#

```
System.bool InsertBasePart(  
    System.string FileName  
)
```

C++/CLI

```
System.bool InsertBasePart(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Visual Basic for Applications (VBA) Syntax

See [PartDoc::InsertBasePart](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertBendNotes Method (IPartDoc)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IPartDoc Interface](#) : InsertBendNotes Method (IPartDoc)

Inserts bend notes for the specified flat-pattern feature of this sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBendNotes( _  
    ByVal FlatPatternFeature As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim FlatPatternFeature As System.Object  
Dim value As System.Object
```

```
value = instance.InsertBendNotes(FlatPatternFeature)
```

C#

```
System.object InsertBendNotes(  
    System.object FlatPatternFeature  
)
```

C++/CLI

```
System.Object^ InsertBendNotes(  
& System.Object^ FlatPatternFeature  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[FlatPatternFeature](#)
[IFeature](#)

Return Value

Array of [INotes](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::InsertBendNotes](#).

Example

'VBA

- ' 1. Open a sheet metal part with a **Flat-Pattern1** feature.
- ' 2. Run the macro.
- ' 3. Inspect the Immediate window for the list of inserted bend notes.

```
Dim swApp As SldWorks.SldWorks  
Dim Part As SldWorks.ModelDoc2  
Dim myFlatPattern As SldWorks.Feature  
Dim myBendNotes As Variant  
Dim myAnno As SldWorks.Note  
Dim boolstatus As Boolean  
Dim i As Long  
Option Explicit  
  
Sub main()  
  
    Set swApp = Application.SldWorks  
    Set Part = swApp.ActiveDoc  
    boolstatus = Part.Extension.SelectByID2("Flat-Pattern1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)  
  
    Set myFlatPattern = Part.SelectionManager.GetSelectedObject6(1, -1)  
    If Not myFlatPattern Is Nothing Then  
        myBendNotes = Part.InsertBendNotes(myFlatPattern)  
        For i = 0 To UBound(myBendNotes)  
            Set myAnno = myBendNotes(i)  
            Debug.Print myAnno.GetName  
        Next i  
    End If  
End Sub
```

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

InsertBends Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : InsertBends Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::InsertBends2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBends( _  
    ByVal Radius As System.Double, _  
    ByVal UseBendTable As System.String, _  
    ByVal UseKfactor As System.Double, _  
    ByVal UseBendAllowance As System.Double, _  
    ByVal UseAutoRelief As System.Boolean, _  
    ByVal OffsetRatio As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Radius As System.Double  
Dim UseBendTable As System.String  
Dim UseKfactor As System.Double  
Dim UseBendAllowance As System.Double  
Dim UseAutoRelief As System.Boolean  
Dim OffsetRatio As System.Double  
Dim value As System.Boolean  
  
value = instance.InsertBends(Radius, UseBendTable, UseKfactor, UseBendAllowance, UseAutoRelief, OffsetRatio)
```

C#

```
System.bool InsertBends(  
    System.double Radius,  
    System.string UseBendTable,  
    System.double UseKfactor,  
    System.double UseBendAllowance,  
    System.bool UseAutoRelief,  
    System.double OffsetRatio  
)
```

C++/CLI

```
System.bool InsertBends(  
& System.double Radius,  
& System.String^ UseBendTable,  
& System.double UseKfactor,  
& System.double UseBendAllowance,  
& System.bool UseAutoRelief,  
& System.double OffsetRatio  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Radius

UseBendTable

UseKfactor

UseBendAllowance

UseAutoRelief

OffsetRatio

Visual Basic for Applications (VBA) Syntax

See [PartDoc.InsertBends](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertBends2 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : InsertBends2 Method (IPartDoc)

Creates bends in a thin-feature part.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBends2( _  
    ByVal Radius As System.Double, _  
    ByVal UseBendTable As System.String, _  
    ByVal UseKfactor As System.Double, _  
    ByVal UseBendAllowance As System.Double, _  
    ByVal UseAutoRelief As System.Boolean, _  
    ByVal OffsetRatio As System.Double, _  
    ByVal DoFlatten As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Radius As System.Double  
Dim UseBendTable As System.String  
Dim UseKfactor As System.Double  
Dim UseBendAllowance As System.Double  
Dim UseAutoRelief As System.Boolean  
Dim OffsetRatio As System.Double  
Dim DoFlatten As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.InsertBends2(Radius, UseBendTable, UseKfactor, UseBendAllowance, UseAutoRelief,  
OffsetRatio, DoFlatten)
```

C#

```
System.bool InsertBends2(  
    System.double Radius,  
    System.string UseBendTable,  
    System.double UseKfactor,  
    System.double UseBendAllowance,  
    System.bool UseAutoRelief,  
    System.double OffsetRatio,  
    System.bool DoFlatten  
)
```

C++/CLI

```
System.bool InsertBends2(  
& System.double Radius,  
& System.String^ UseBendTable,  
& System.double UseKfactor,  
& System.double UseBendAllowance,  
& System.bool UseAutoRelief,  
& System.double OffsetRatio,  
& System.bool DoFlatten  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Radius

Radius of the bends

UseBendTable

Bend table name (.btl file)

UseKfactor

K-Factor ratio or -1 if not used

UseBendAllowance

Bend allowance value or -1 if not used

UseAutoRelief

True if auto-relief cuts are to be added, false if not

OffsetRatio

Distance relief cut extends beyond bend (see **Remarks**)

DoFlatten

True to create these three features: Sheet-Metal, Flatten-Bends, and Process-Bends, false to create only the Sheet-Metal feature

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc.InsertBends2](#).

Remarks

The offsetRatio argument is from 0.05 to 2.0. Any other value fails to create the bend features.

When True is passed to doFlatten, all three Sheet-Metal features are created.

For more information on these arguments, see SOLIDWORKS Help.

See Also

[IPartDoc Interface](#)

[IPartDoc Members](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertBendTable Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : InsertBendTable Method (IPartDoc)

Creates a bend table annotation for the flat pattern of this sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBendTable( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal StartValue As System.String, _  
    ByVal TableTemplate As System.String _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IPartDoc  
Dim X As System.Double  
Dim Y As System.Double
```



```
Dim StartValue As System.String
Dim TableTemplate As System.String
Dim value As System.Object

value = instance.InsertBendTable(X, Y, StartValue, TableTemplate)
```

C#

```
System.Object InsertBendTable(
    System.double X,
    System.double Y,
    System.string StartValue,
    System.string TableTemplate
)
```

C++/CLI

```
System.Object^ InsertBendTable(
    & System.double X,
    & System.double Y,
    & System.String^ StartValue,
    & System.String^ TableTemplate
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X-coordinate for placement of the bend table

Y
Y-coordinate for placement of the bend table

StartValue
Starting datum tag; a value from A to Z for letter tags; a positive integer for number tags

TableTemplate
Full pathname of the template (e.g., *install_dir\lang\language\bendtable-standard.sldbndtbt*)

Return Value

[IBendTableAnnotation](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::InsertBendTable](#).

Example

'VBA

- '1. Ensure the specified part and table template exist.
- '2. Run the macro.
- '3. Inserts a bend table annotation for the flat pattern.
- '4. Inspect the graphics area and the Immediate window.

```
Dim swApp As SldWorks.SldWorks
Dim myBendTableAnno As SldWorks.BendTableAnnotation
Dim myBendTable As SldWorks.BendTable
Dim Part As SldWorks.ModelDoc2
Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long
Option Explicit
Sub main()
```

```
    Set swApp = Application.SldWorks
    Set Part = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS
2019\samples\tutorial\api\SMGussetAPI.SLDPRT", 1, 0, "", longstatus, longwarnings)
    swApp.ActivateDoc2 "SMGussetAPI.SLDPRT", False, longstatus
    Set Part = swApp.ActiveDoc
```

```
    Set myBendTableAnno = Part.InsertBendTable(-4.06616963665726E-02, 6.09432383467686E-02, "A",
"install_dir\lang\english\bendtable-standard.sldbndtbt")
    Set myBendTable = myBendTableAnno.BendTable
    Debug.Print "Tag style of the bend table as defined in swBendTableTagStyle_e: " & myBendTable.TagStyle
```

```
End Sub
```

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IBendTable Interface](#)
[IView::InsertBendTable Method \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertImportedFeature Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : InsertImportedFeature Method (IPartDoc)

Inserts a third-party native CAD file into this part.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertImportedFeature( _  
    ByVal FileName As System.String, _  
    ByRef Errors As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim FileName As System.String  
Dim Errors As System.Integer  
Dim value As System.Object  
  
value = instance.InsertImportedFeature(FileName, Errors)
```

C#

```
System.object InsertImportedFeature(  
    System.string FileName,  
    out System.int Errors  
)
```

C++/CLI

```
System.Object^ InsertImportedFeature(  
& System.String^ FileName,  
& [Out] System.int Errors  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName
Full path name of the third-party native CAD file to insert
Errors
Error code as defined in [swFileLoadError_e](#)

Return Value

[IFeature](#) if successful; null or Nothing otherwise

Visual Basic for Applications (VBA) Syntax

See [PartDoc::InsertImportedFeature](#).

Remarks

To successfully use this method, you must first turn on 3D Interconnect by either:

- Setting **Tools > Options > System Options > Import > Enable 3D Interconnect**
- OR -
- Calling [ISldWorks::SetUserPreferenceToggle\(swUserPreferenceToggle_e.swMultiCAD_Enable3DInterconnect, True\)](#)

The feature is imported and positioned relative to the origin using coordinates of the imported file. To edit the feature, use [IFeature::GetImportedFeatureParameters](#) and [IFeature::SetImportedFeatureParameters](#).

See the [IImport3DInterconnectData](#) Remarks.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[ISldWorks::LoadFile4 Method \(\)](#)
[IAssemblyDoc::InsertImportedComponent Method \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMirrorAll Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : InsertMirrorAll Method (IPartDoc)

Mirrors the part about a selected planar face.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMirrorAll() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.Boolean

value = instance.InsertMirrorAll()
```

C#

```
System.bool InsertMirrorAll()
```

C++/CLI

```
System.bool InsertMirrorAll();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the feature is successfully created, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::InsertMirrorAll](#).

Remarks

This method returns a True or false to indicate whether or not the mirror-all feature was created. If it is successful, the newly created feature remains selected after the method runs. You can use [ISelectionMgr::GetSelectedObject6](#) to retrieve this object.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

[IPartDoc::MirrorPart Method](#)
[IFeatureManager::InsertMirrorFeature Method](#)
[IPartDoc::MirrorFeature Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMirrorFeature Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : InsertMirrorFeature Method (IPartDoc)

Obsolete. Superseded by [IFeatureManager::InsertMirrorFeature](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertMirrorFeature( _  
    ByVal GeometryPattern As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim GeometryPattern As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.InsertMirrorFeature(GeometryPattern)
```

C#

```
System.bool InsertMirrorFeature(  
    System.bool GeometryPattern  
)
```

C++/CLI

```
System.bool InsertMirrorFeature(  
    & System.bool GeometryPattern  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GeometryPattern

Visual Basic for Applications (VBA) Syntax

See [PartDoc::InsertMirrorFeature](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertPart Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : InsertPart Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::InsertPart2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertPart( _
    ByVal FileName As System.String, _
    ByVal ImportPlane As System.Boolean, _
    ByVal ImportAxis As System.Boolean, _
    ByVal ImportCThread As System.Boolean _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim FileName As System.String
Dim ImportPlane As System.Boolean
Dim ImportAxis As System.Boolean
Dim ImportCThread As System.Boolean
Dim value As Feature

value = instance.InsertPart(FileName, ImportPlane, ImportAxis, ImportCThread)
```

C#

```
Feature InsertPart(
    System.string FileName,
    System.bool ImportPlane,
    System.bool ImportAxis,
    System.bool ImportCThread
)
```

C++/CLI

```
Feature^ InsertPart(
    & System.String^ FileName,
    & System.bool ImportPlane,
    & System.bool ImportAxis,
    & System.bool ImportCThread
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Name of part file

ImportPlane

True if the planes from the part should be imported into this part, false if not

ImportAxis

True if the axes from the part should be imported into this part, false if not

ImportCThread

True if the cosmetic threads from the part should be imported into this part, false if not

Return Value

New [feature](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::InsertPart](#).

Remarks

This method inserts the specified part at the origin of this part. To position the insert part at a different location or orientation or both, use [IFeatureManager::InsertMoveCopyBody2](#).

The interface returned by this method is [IFeature](#), which gives you access to the feature APIs such as [IFeature::Name](#) to get or set the name of the feature.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertPart2 Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : InsertPart2 Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::InsertPart3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertPart2( _  
    ByVal FileName As System.String, _  
    ByVal Options As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim FileName As System.String  
Dim Options As System.Integer  
Dim value As Feature  
  
value = instance.InsertPart2(FileName, Options)
```

C#

```
Feature InsertPart2(  
    System.string FileName,  
    System.int Options  
)
```

C++/CLI

```
Feature^ InsertPart2(  
& System.String^ FileName,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Name of part file

Options

Insertion options as defined in [swInsertPartOptions_e](#)

Return Value

Inserted [feature](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::InsertPart2](#).

Remarks

This method inserts the specified part at the origin of this part. To position the inserted part at a different location or orientation or both, use [IFeatureManager::InsertMoveCopyBody2](#).

See Also

[IPartDoc Interface](#)

[IPartDoc Members](#)

[IDerivedPartFeatureData Interface](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertPart3 Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : InsertPart3 Method (IPartDoc)

Inserts the specified part in the specified configuration into this part.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function InsertPart3( _  
    ByVal FileName As System.String, _  
    ByVal Options As System.Integer, _  
    ByVal ConfigurationName As System.String _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim FileName As System.String  
Dim Options As System.Integer  
Dim ConfigurationName As System.String  
Dim value As Feature  
  
value = instance.InsertPart3(FileName, Options, ConfigurationName)
```

C#

```
Feature InsertPart3(  
    System.string FileName,  
    System.int Options,  
    System.string ConfigurationName  
)
```

C++/CLI

```
Feature^ InsertPart3(  
& System.String^ FileName,  
& System.int Options,  
& System.String^ ConfigurationName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Name of the part file to insert

Options

Insertion options as defined in [swInsertPartOptions_e](#)

ConfigurationName

Name of FileName's configuration

Return Value

Inserted [feature](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::InsertPart3](#).

Example

[Modify Derived Part \(VBA\)](#)
[Modify Derived Part \(VB.NET\)](#)
[Modify Derived Part \(C#\)](#)

Remarks

This method inserts the specified part at the origin of this part. To position the inserted part at a different location or orientation or both, use [IFeatureManager::InsertMoveCopyBody2](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IDerivedPartFeatureData Interface](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISetEntityName Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ISetEntityName Method (IPartDoc)

Sets the name of the entity if the entity does not have a name.

.NET Syntax

Visual Basic (Declaration)

```
Function ISetEntityName( _  
    ByVal Entity As Entity, _  
    ByVal StringValue As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Entity As Entity  
Dim StringValue As System.String  
Dim value As System.Boolean  
  
value = instance.ISetEntityName(Entity, StringValue)
```

C#

```
System.bool ISetEntityName(  
    Entity Entity,  
    System.string StringValue  
)
```

C++/CLI

```
System.bool ISetEntityName(  
& Entity^ Entity,  
& System.String^ StringValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entity
[Entity](#)
StringValue
Name of the entity

Return Value

True if name is set successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ISetEntityName](#).

Remarks

If the entity already has a name, then this method does not change the name and returns false.

This behavior is intended to prevent a program from accidentally renaming an entity that is referenced in some other location. For example, if an assembly contains a mate to a face on a part, then a name is automatically assigned to that face. If you were to change that name, then there is no guarantee that the mate is still valid. Therefore, when using entity names, you should first check to see if the entity is already named, and, if so, use the existing name. If no name exists for the entity, then you can name the entity.

You can explicitly delete an entity name using the [IPartDoc::DeleteEntityName](#) or [IPartDoc::!DeleteEntityName](#) method. You then have the option of renaming the item or using that name elsewhere. The method was provided because the action is already available in the core SOLIDWORKS Design software. However, you should recognize the possibility of reference failures as described.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::SetEntityName Method](#)
[IPartDoc::GetEntityByName Method](#)
[IPartDoc::GetEntityName Method](#)
[IPartDoc::GetNamedEntities Method](#)
[IPartDoc::GetNamedEntitiesCount Method](#)
[IPartDoc::!GetEntityByName Method](#)
[IPartDoc::!GetEntityName Method](#)
[IPartDoc::!GetNamedEntities Method](#)
[IPartDoc::DeleteEntityName Method](#)
[IPartDoc::!DeleteEntityName Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsMirrored Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IsMirrored Method (IPartDoc)

Gets whether this part is mirrored.

.NET Syntax

Visual Basic (Declaration)

```
Function IsMirrored() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.Boolean  
  
value = instance.IsMirrored()
```

C#

```
System.bool IsMirrored()
```

C++/CLI

```
System.bool IsMirrored();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the part is mirrored, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IsMirrored](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IComponent2::IsMirrored Method](#)
[IAssemblyDoc::MirrorComponents3 Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsWeldment Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : IsWeldment Method (IPartDoc)

Gets whether this part contains a weldment feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IsWeldment() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.Boolean

value = instance.IsWeldment()
```

C#

```
System.bool IsWeldment()
```

C++/CLI

```
System.bool IsWeldment();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if part contains a weldment feature, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::IsWeldment](#).

Example

[Does Part Contain a Weldment Feature \(VBA\)](#)

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

Availability

SOLIDWORKS 2006 SP2, Revision Number 14.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

MakeSection Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : MakeSection Method (IPartDoc)

Saves sketch profiles for use in the wizard.

.NET Syntax

Visual Basic (Declaration)

Sub MakeSection()

Visual Basic (Usage)



```
Dim instance As IPartDoc  
  
instance.MakeSection()
```

C#

```
void MakeSection()
```

C++/CLI

```
void MakeSection();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::MakeSection](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

MaterialIdName Property (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : MaterialIdName Property (IPartDoc)

Gets or sets the material name.

.NET Syntax

Visual Basic (Declaration)

```
Property MaterialIdName As System.String
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim value As System.String  
  
instance.MaterialIdName = value  
  
value = instance.MaterialIdName
```

C#

```
System.string MaterialIdName {get; set;}
```

C++/CLI

```
property System.String^ MaterialIdName {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Material name (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::MaterialIdName](#).

Example

[Get Material Property Names \(VBA\)](#)
[Set Material Property Name \(VBA\)](#)

Remarks

This property returns a string that contains three pieces of information separated by a pipe "|":

- Database name
- Material name
- Database ID of material

Example:

MY MATERIALS|1.0038 (S235JR)|277

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetMaterialPropertyName2 Method](#)
[IPartDoc::GetMaterialVisualProperties Method](#)
[IPartDoc::SetMaterialPropertyName2 Method](#)
[IPartDoc::SetMaterialVisualProperties Method](#)
[IPartDoc::MaterialPropertyValues Property](#)
[IPartDoc::MaterialPropertyValues Property](#)
[IPartDoc::MaterialUserName Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

MaterialPropertyValues Property (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : MaterialPropertyValues Property (IPartDoc)

Gets or sets a material's properties in the active configuration.

.NET Syntax

Visual Basic (Declaration)

Property MaterialPropertyValues As System.Object

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.Object

instance.MaterialPropertyValues = value
value = instance.MaterialPropertyValues
```

C#

```
System.Object MaterialPropertyValues {get; set;}
```

C++/CLI

```
property System.Object^ MaterialPropertyValues {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles describing the material's values (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::MaterialPropertyValues](#).

Remarks

The material values include the color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency and emission.

The format of the parameters or return values is an array of doubles as follows:

[*R, G, B, Ambient, Diffuse, Specular, Shininess, Transparency, Emission*]

To see a color change, you must:

1. Specify *R*, *G*, and *B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse*, *Specular*, *Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient*, *Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after setting this property.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::MaterialPropertyValues Property](#)
[IPartDoc::MaterialIdName Property](#)
[IPartDoc::MaterialUserName Property](#)
[IPartDoc::SetMaterialPropertyName2 Method](#)
[IPartDoc::SetMaterialVisualProperties Method](#)
[IPartDoc::GetMaterialPropertyName2 Method](#)
[IPartDoc::GetMaterialVisualProperties Method](#)

SOLIDWORKS API Help

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MaterialUserName Property (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : MaterialUserName Property (IPartDoc)

Gets or sets the material name. This name is visible to the user.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property MaterialUserName As System.String
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
Dim value As System.String

instance.MaterialUserName = value

value = instance.MaterialUserName
```

C#

```
System.string MaterialUserName {get; set;}
```

C++/CLI

```
property System.String^ MaterialUserName {
    System.String^ get();
    void set ( & System.String^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Material's user name property

Visual Basic for Applications (VBA) Syntax

See [PartDoc::MaterialUserName](#).

Example

[Get Material Property Names \(VBA\)](#)

Remarks

This name is visible to the user.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::MaterialIdName Property](#)
[IPartDoc::MaterialPropertyValues Property](#)
[IPartDoc::IMaterialPropertyValues Property](#)
[IPartDoc::SetMaterialVisualProperties Method](#)
[IPartDoc::SetMaterialPropertyName2 Method](#)
[IPartDoc::GetMaterialVisualProperties Method](#)
[IPartDoc::GetMaterialPropertyName2 Method](#)

SOLIDWORKS API Help

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MirrorFeature Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : MirrorFeature Method (IPartDoc)

Displays a dialog that allows the end-user to mirror a feature about a selected plane or planar face.

.NET Syntax

Visual Basic (Declaration)

```
Sub MirrorFeature()
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
```

```
instance.MirrorFeature()
```

C#

```
void MirrorFeature()
```

C++/CLI

```
void MirrorFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::MirrorFeature](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::InsertMirrorAll Method](#)
[IPartDoc::MirrorPart Method](#)
[IFeatureManager::InsertMirrorFeature Method](#)

SOLIDWORKS API Help

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MirrorPart Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : MirrorPart Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::MirrorPart2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub MirrorPart()
```

```
Dim instance As IPartDoc

instance.MirrorPart()
```

C#

```
void MirrorPart()
```

C++/CLI

```
void MirrorPart();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::MirrorPart](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::InsertMirrorAll Method](#)
[IPartDoc::MirrorFeature Method](#)
[IFeatureManager::InsertMirrorFeature Method](#)

SOLIDWORKS API Help

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MirrorPart2 Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : MirrorPart2 Method (IPartDoc)

Creates a new part document that mirrors this part about a selected reference plane or planar face.

.NET Syntax

Visual Basic (Declaration)

```
Function MirrorPart2( _
    ByVal BreakLink As System.Boolean, _
    ByVal Options As System.Integer, _
    ByRef ResultPart As ModelDoc2 _
) As Feature
```

Visual Basic (Usage)

Copy Code

```
Dim instance As IPartDoc
Dim BreakLink As System.Boolean
Dim Options As System.Integer
Dim ResultPart As ModelDoc2
Dim value As Feature

value = instance.MirrorPart2(BreakLink, Options, ResultPart)
```

C#

```
Feature MirrorPart2(
    System.bool BreakLink,
    System.int Options,
    out ModelDoc2 ResultPart
)
```

C++/CLI

```
Feature^ MirrorPart2(
    & System.bool BreakLink,
    & System.int Options,
```

& [Out] ModelDoc2^ [ResultPart](#)
)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BreakLink

True to break the link to the original part, false to not

Options

Insertion options as defined in [swMirrorPartOptions](#) e

ResultPart

[IModelDoc2](#)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::MirrorPart2](#).

Example

[Get Mirror Part Information \(VBA\)](#)
[Get Mirror Part Information \(VB.NET\)](#)
[Get Mirror Part Information \(C#\)](#)
[Mirror Sheet-metal Part \(C#\)](#)
[Mirror Sheet-metal Part \(VB.NET\)](#)
[Mirror Sheet-metal Part \(VBA\)](#)

Remarks

Before calling this method, you must select a reference plane or planar face about which to mirror this part.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IMirrorPartFeatureData Interface](#)
[IPartDoc::InsertMirrorAll Method \(\)](#)
[IPartDoc::MirrorFeature Method \(\)](#)
[IFeatureManager::InsertMirrorFeature2](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveAllDisplayStates Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : RemoveAllDisplayStates Method (IPartDoc)

Removes all display states and appearances from this part document.

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveAllDisplayStates() As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IPartDoc  
Dim value As System.Boolean
```

```
value = instance.RemoveAllDisplayStates()
```

C#

```
System.bool RemoveAllDisplayStates()
```

C++/CLI



```
System.bool RemoveAllDisplayStates();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::RemoveAllDisplayStates](#).

Example

[Clear Display States \(VBA\)](#)
[Clear Display States \(VB.NET\)](#)
[Clear Display States \(C#\)](#)

Remarks

Call this method to:

- Remove all appearances from all configurations of this part.
- Remove all display states except the active display state.
- Un-link display states from configurations.
- Show all hidden features.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IConfigurationManager::LinkDisplayStatesToConfigurations Property \(\)](#)

Availability

SOLIDWORKS 2011 SP4, Revision Number 19.4

SOLIDWORKS API Help

[Send comments](#) on this topic.

ReorderFeature Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ReorderFeature Method (IPartDoc)

Reorders features and their operations.

.NET Syntax

Visual Basic (Declaration)

```
Function ReorderFeature( _  
    ByVal FeatureToMove As System.String, _  
    ByVal MoveAfterFeature As System.String _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IPartDoc  
Dim FeatureToMove As System.String  
Dim MoveAfterFeature As System.String  
Dim value As System.Boolean
```

```
value = instance.ReorderFeature(FeatureToMove, MoveAfterFeature)
```

C#

```
System.bool ReorderFeature(  
    System.string FeatureToMove,  
    System.string MoveAfterFeature  
)
```



```
System.bool ReorderFeature(
& System.String^ FeatureToMove,
& System.String^ MoveAfterFeature
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatureToMove

Name of the feature to move

MoveAfterFeature

Name of the feature that now precedes the feature in the FeatureManager design tree

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ReorderFeature](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::FirstFeature Method](#)
[IAssemblyDoc::ReorderComponents Method](#)
[IModelDocExtension::ReorderFeature Method \(\)](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ResetPropertyExtension Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : ResetPropertyExtension Method (IPartDoc)

Clears all values stored in the property extension.

.NET Syntax

Visual Basic (Declaration)

```
Sub ResetPropertyExtension()
```

Visual Basic (Usage)

```
Dim instance As IPartDoc
instance.ResetPropertyExtension()
```

C#

```
void ResetPropertyExtension()
```

C++/CLI

```
void ResetPropertyExtension();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::ResetPropertyExtension](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::AddPropertyExtension Method](#)
[IPartDoc::GetPropertyExtension Method](#)

SaveToFile Method (IPartDoc)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : SaveToFile Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::SaveToFile2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SaveToFile()
```

Visual Basic (Usage)



```
Dim instance As IPartDoc
```

```
instance.SaveToFile()
```

C#

```
void SaveToFile()
```

C++/CLI

```
void SaveToFile();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [PartDoc::SaveToFile](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

SaveToFile2 Method (IPartDoc)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : SaveToFile2 Method (IPartDoc)

Obsolete. Superseded by [IPartDoc::SaveToFile3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SaveToFile2( _  
    ByVal Name As System.String, _  
    ByVal Options As System.Integer, _  
    ByRef Errors As System.Integer, _  
    ByRef Warnings As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Name As System.String  
Dim Options As System.Integer  
Dim Errors As System.Integer  
Dim Warnings As System.Integer  
Dim value As System.Boolean
```

```
value = instance.SaveToFile2(Name, Options, Errors, Warnings)
```

C#

```
System.bool SaveToFile2(  
    System.string Name,  
    System.int Options,  
    out System.int Errors,  
    out System.int Warnings  
)
```

C++/CLI

```
System.bool SaveToFile2(  
& System.String^ Name,  
& System.int Options,  
& [Out] System.int Errors,  
& [Out] System.int Warnings  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Name of the part document (.sldprt)

Options

Relevant options indicating how to save the document as defined in [swSaveAsOptions_e](#)

Errors

Errors that caused the save to fail as defined in [swFileSaveError_e](#)

Warnings

Warnings or extra information generated during the save operation as defined in [swFileSaveWarning_e](#)

Return Value

True if the save is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::SaveToFile2](#).

See Also

[IPartDoc Interface](#)

[IPartDoc Members](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

SaveToFile3 Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : SaveToFile3 Method (IPartDoc)

Saves the selected weldment member, surface body, or solid body to another part document.

.NET Syntax

Visual Basic (Declaration)

```
Function SaveToFile3(  
    ByVal Name As System.String, _  
    ByVal Options As System.Integer, _  
    ByVal CutListProps As System.Integer, _  
    ByVal OverrideTemplate As System.Boolean, _  
    ByVal TemplatePath As System.String, _  
    ByRef Errors As System.Integer, _  
    ByRef Warnings As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```

Dim instance As IPartDoc
Dim Name As System.String
Dim Options As System.Integer
Dim CutListProps As System.Integer
Dim OverrideTemplate As System.Boolean
Dim TemplatePath As System.String
Dim Errors As System.Integer
Dim Warnings As System.Integer
Dim value As System.Boolean

```

```

value = instance.SaveToFile3(Name, Options, CutListProps, OverrideTemplate, TemplatePath, Errors, Warnings)

```

C#

```

System.bool SaveToFile3(
    System.string Name,
    System.int Options,
    System.int CutListProps,
    System.bool OverrideTemplate,
    System.string TemplatePath,
    out System.int Errors,
    out System.int Warnings
)

```

C++/CLI

```

System.bool SaveToFile3(
    & System.String^ Name,
    & System.int Options,
    & System.int CutListProps,
    & System.bool OverrideTemplate,
    & System.String^ TemplatePath,
    & [Out] System.int Errors,
    & [Out] System.int Warnings
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Path and name of the part document to which to save the selected weldment member, solid body, or surface body

Options

How to save the document as defined in [swSaveAsOptions_e](#)

CutListProps

Option for transferring the cut list of the selected weldment member, solid body, or surface body to the new part as defined in [swCutListTransferOptions_e](#)

OverrideTemplate

True to override the part template with the template specified by TemplatePath, false to not

TemplatePath

Path to part template; valid only if OverrideTemplate is true

Errors

Save errors as defined in [swFileSaveError_e](#)

Warnings

Warnings or extra information generated during the save operation as defined in [swFileSaveWarning_e](#)

Return Value

True if the save is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::SaveToFile3](#).

Example

[Save Solid Body to File \(VBA\)](#)
[Save Solid Body to File \(VB.NET\)](#)
[Save Solid Body to File \(C#\)](#)

Remarks

The difference between this method and the now obsolete IPartDoc::SaveToFile2 is that this method allows you to specify in CutListProps how the cut list properties of the selected weldment member of a weldment part are transferred to the saved part.

Before calling this method, select a weldment member, surface body, or solid body in the FeatureManager design tree.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetCosmosWorksMaterial Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : SetCosmosWorksMaterial Method (IPartDoc)

Assigns the material to use during analysis to this part.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetCosmosWorksMaterial( _  
    ByVal ConfigName As System.String, _  
    ByVal Type As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim ConfigName As System.String  
Dim Type As System.Integer
```

```
instance.SetCosmosWorksMaterial(ConfigName, Type)
```

C#

```
void SetCosmosWorksMaterial(  
    System.string ConfigName,  
    System.int Type  
)
```

C++/CLI

```
void SetCosmosWorksMaterial(  
& System.String^ ConfigName,  
& System.int Type  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName

Name of the configuration

Type

Type of material to assign as defined by [swCosmosWorksMat](#)

Return Value

True if the material is assigned, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::SetCosmosWorksMaterial](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetDroppedFileName Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : SetDroppedFileName Method (IPartDoc)

Sets the filename for a recently dropped file.

.NET Syntax

Visual Basic (Declaration)

```
Function SetDroppedFileName( _  
    ByVal FileName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim FileName As System.String  
Dim value As System.Boolean  
  
value = instance.SetDroppedFileName(FileName)
```

C#

```
System.bool SetDroppedFileName(  
    System.string FileName  
)
```

C++/CLI

```
System.bool SetDroppedFileName(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Name of the file to use for the rest of the drop process

Return Value

True if name is set, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::SetDroppedFileName](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IAssemblyDoc::SetDroppedFileName Method](#)
[DPartDocEvents FileDropPreNotifyEventHandler Delegate](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetEntityName Method (IPartDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : SetEntityName Method (IPartDoc)

Sets the name of the entity if the entity does not have a name and the name is unique to the part.

.NET Syntax

Visual Basic (Declaration)

```
Function SetEntityName( _  
    ByVal Entity As System.Object, _  
    ByVal StringValue As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Entity As System.Object  
Dim StringValue As System.String  
Dim value As System.Boolean  
  
value = instance.SetEntityName(Entity, StringValue)
```

C#

```
System.bool SetEntityName(  
    System.object Entity,  
    System.string StringValue  
)
```

C++/CLI

```
System.bool SetEntityName(  
& System.Object^ Entity,  
& System.String^ StringValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entity

[Entity](#)

StringValue

Name of the entity

Return Value

True if name is set successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [PartDoc::SetEntityName](#).

Example

[Select and Rename Object \(VBA\)](#)

[Get Entity By Name \(VBA\)](#)

[Get Cross Break Feature Data in Sheet Metal Part \(C#\)](#)

[Get Cross Break Feature Data in Sheet Metal Part \(VB.NET\)](#)

[Get Cross Break Feature Data in Sheet Metal Part \(VBA\)](#)

Remarks

If the entity already has a name or the specified name is not unique to the part, then this method does not change the name and returns false.

This behavior is intended to prevent a program from accidentally renaming an entity that is referenced in some other location. For example, if an assembly contains a mate to a face on a part, then a name is automatically assigned to that face. If you were to change that name, then there is no guarantee that the mate is still valid. Therefore, when using entity names, you should first check to see if the entity is already named, and, if so, use the existing name. If no name exists for the entity, then you can name the entity.

You can explicitly delete an entity name using the [IPartDoc::DeleteEntityName](#) or [IPartDoc::!DeleteEntityName](#) method. You then have the option of renaming the item or using that name elsewhere. The method was provided because the action is already available in the core SOLIDWORKS Design software. However, you should recognize the possibility of reference failures as described.

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::GetEntityByName Method](#)
[IPartDoc::GetEntityName Method](#)
[IPartDoc::IGetEntityByName Method](#)
[IPartDoc::IGetEntityName Method](#)
[IPartDoc::IGetNamedEntities Method](#)
[IPartDoc::GetNamedEntities Method](#)
[IPartDoc::GetNamedEntitiesCount Method](#)
[IPartDoc::ISetEntityName Method](#)
[IPartDoc::IDeleteEntityName Method](#)
[IPartDoc::DeleteEntityName Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetLineColor Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : SetLineColor Method (IPartDoc)

Sets the color for the lines in the part document.

Visual Basic Syntax

Visual Basic (Declaration)

```
Sub SetLineColor( _  
    ByVal Color As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Color As System.Integer  
  
instance.SetLineColor(Color)
```

C#

```
void SetLineColor(  
    System.int Color  
)
```

C++/CLI

```
void SetLineColor(  
& System.int Color  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Color

Color definition as a COLORREF

Visual Basic for Applications (VBA) Syntax

See [PartDoc::SetLineColor](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::SetLineStyle Method](#)
[IPartDoc::SetLineWidth Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SetLineStyle Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : SetLineStyle Method (IPartDoc)

Sets the style or font for the lines in the part document.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetLineStyle( _  
    ByVal StyleName As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim StyleName As System.String  
  
instance.SetLineStyle(StyleName)
```

C#

```
void SetLineStyle(  
    System.string StyleName  
)
```

C++/CLI

```
void SetLineStyle(  
& System.String^ StyleName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StyleName

Style or font for the line as defined in [swLineStyles_e](#)

Visual Basic for Applications (VBA) Syntax

See [PartDoc::SetLineStyle](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::SetLineWidth Method](#)
[IPartDoc::SetLineColor Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SetLineWidth Method (IPartDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IPartDoc Interface](#) : SetLineWidth Method (IPartDoc)

Sets the thickness or weight for the lines in the part document.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetLineWidth(  
    ByVal Width As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IPartDoc  
Dim Width As System.Integer  
  
instance.SetLineWidth(Width)
```

C#

```
void SetLineWidth(  
    System.int Width  
)
```

C++/CLI

```
void SetLineWidth(  
& System.int Width  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Width

Line thickness or weight as defined in [swLineWeights](#) e

Visual Basic for Applications (VBA) Syntax

See [PartDoc::SetLineWidth](#).

See Also

[IPartDoc Interface](#)
[IPartDoc Members](#)
[IPartDoc::SetLineColor Method](#)
[IPartDoc::SetLineStyle Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0